

Course Code: CS2228		
Name of the Programme: B.Tech (Hons.) CSE		
Name of the Course: Deep Learning		
Semester: IV		
Course Credits:	No. of hours per week (L+T+P):	Total no. of Teaching hours:
3	2+0+2	60
Course Objectives		
A. To introduce the foundational concepts of neural networks and their application in classification tasks.		
B. To explore advanced deep learning architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for solving computer vision and sequence modeling problems.		
C. To provide hands-on experience in building, training, and evaluating deep learning models using real-world datasets.		
D. To apply explainable AI techniques and industry-standard tools to interpret, optimize, and deploy deep learning models on various platforms.		

Syllabus:

Module – 1: Introduction to Deep Learning	No. of hours
	6
Overview of machine learning and deep learning, History and Evolution of Deep Learning. Introduction to Neural networks: Perceptron, Multilayer Perceptron, Backpropagation, Shallow neural networks, deep neural networks, Optimizers: Gradient Descent (GD), Momentum-Based GD, Nesterov Accelerated GD, Stochastic GD, AdaDelta, AdaGrad, RMSProp, Adam, Loss Functions. Introduction to CUDA and GPU.	

Module – 2: Convolutional Neural Networks	No. of hours
	6
Introduction to Convolutional Neural Networks, Layers in CNN, Types of convolutions, Regularization Techniques: L1/L2 regularization, dropout, Early stopping, Data augmentation, A typical CNN structure, Standard CNN models: AlexNet, VGGNet, GoogLeNet, ResNet, Inception.	

Module - 3: Unsupervised Learning and Generative Modeling with Autoencoders and GANs	No. of hours
	6
Autoencoder (Types: Linear, CNN-based), Training Autoencoders, Variational Autoencoders (VAE), Introduction to GAN, GAN Architecture (Generator, Discriminator), Applications.	

Module - 4: Unsupervised Learning and Generative Modeling with	No. of hours
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Autoencoders and GANs	6
Sequence-to-sequence models: Introduction to Recurrent Neural Networks and their structure, Challenges in RNN (Vanishing and Exploding Gradients), Long-short term memory (LSTM), Gated Recurrent Unit (GRU), Attention mechanism in RNNs, and Transformers.	

Module - 5: Model Deployment with Industry SDKs	No. of hours
	6
Overview of deep learning SDKs, Importance of optimization and deployment in deep learning, Intel's OpenVINO Toolkit, Qualcomm's SNPE, NVIDIA TensorRT.	

Course Outcomes: After completing the course, the students will be able to	
a	Apply neural networks to perform classification tasks.
b	Implement Convolutional Neural Networks (CNNs) for image classification.
c	Apply VAE and GAN for generating the data, particularly for image data..
d	Implement sequence modeling and deploy optimized deep learning models using industry-standard SDKs such as OpenVINO, SNPE, and TensorRT.

Text Books	
1	I. Goodfellow, Y. Bengio, and A. Courville, <i>Deep Learning</i> . Cambridge, MA: MIT Press, 2016. ISBN: 9780262035613.
2	A. Zhang <i>et al.</i> , <i>Dive into Deep Learning</i> . Cambridge, UK: Cambridge University Press, 2023. ISBN: 9781009389433.

Reference Books	
1	M. Nielsen, <i>Neural Networks and Deep Learning</i> , 2019. [Online]. Available: http://neuralnetworksanddeeplearning.com
2	A. Géron, <i>Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow</i> , 3rd ed. Shroff/O'Reilly, 2019. ISBN: 1492032646.

Lab Programs		[30 Hours]
1	Write a program to implement a classification problem using an Artificial Neural Network (ANN). (a) Build the model using varying numbers of hidden layers and neurons in each hidden layer. (b) Use different optimizers and compare their accuracies. (c) Apply dropout and compare the results with those obtained without using dropout by varying the dropout rate. (d) Plot graphs to visualize all the results obtained from the above steps (e) Analyze the impact of hyperparameter tuning (e.g., learning rate, batch size) on model performance. (f) Interpret the results using LIME and SHAP	
2	Create a program to implement a Convolutional Neural Network (CNN) for the given dataset. (a) Design a customized CNN model. (b) Experiment with and without Batch Normalization and compare their accuracy. (c) Evaluate the results against at least three different CNN variants. (d) Visualize the results by plotting graphs (e) Investigate techniques for handling potential dataset imbalances or applying advanced data augmentation	
3	Implement a program to build Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) models for the given dataset. (a) Create customized RNN and LSTM architectures. (b) Experiment with variants of RNN and LSTM and compare their performance in terms of accuracy. (c) Analyze and compare the results obtained from RNN and LSTM models. (d) Plot graphs to visualize the loss and accuracy metrics.	
4	Develop a program to implement a Convolutional Neural Network (CNN) for the given dataset. (a) Construct the CNN model and train it on the dataset. (b) Evaluate the model performance and interpret the results using explainable AI techniques: LIME, SHAP, SmoothGRAD, Occlusion, Saliency Maps (c) Visualize the interpretations through graphical representations.	
5	Speed up deep learning inference using NVIDIA TensorRT - Classification	